

CSCE 632 ACCESSIBLE COMPUTING · MODULE 03 · LECTURE NOTES

Vision

No sight, low sight, and colour — three different requirements.

MODULE OUTCOMES

- Distinguish the three vision requirements.
- Explain the screen reader navigation model.
- State the colour and contrast requirements.
- Explain magnification, zoom, and reflow.
- Test an interface against all three.

1 Three requirements

NO VISION, LOW VISION, AND COLOUR VISION ARE THREE SEPARATE REQUIREMENTS WITH DIFFERENT FIXES

No vision needs a non-visual representation of the whole interface — correct semantics, so that a screen reader can convey structure, state, and change (§2 and Module 08). Visual design is entirely irrelevant to it: a beautiful interface and an ugly one are identical here.

Low vision needs the visual design to survive enlargement — magnification, browser zoom, reflow, and contrast (§§3 and 4). Semantics help and are not sufficient, because the user is looking at the pixels.

Colour vision needs redundant encoding — colour must never be the only channel carrying information (§3) — which is a single rule, is trivially testable, and is very widely broken.

And the three are frequently conflated — 'accessible for blind users' is used loosely to mean all three, and a fix for one does nothing at all for the others: perfect semantics do not fix low contrast, and a contrast fix does not make anything announceable. **That conflation is how interfaces end up half-done and reported as finished.**

2 The screen reader model

NOT linearly, usually. The common operations:

jump by heading	the primary navigation, which is why heading structure IS navigation
jump by landmark	main, nav, search, footer
list all links	then filter by typing
list all headings	to get an outline of the page
tab through controls	focusable things only
read continuously	when actually reading prose
table navigation	cell by cell, with the row and column headers announced for each cell

WHICH MEANS

headings are a table of contents, not a font size
 a page with one heading has no navigation
 link text must make sense out of context, because
 it is routinely heard out of context
 and "click here" twelve times is a list of twelve
 identical items

Heading structure is navigation, not typography — which is the single most useful thing to know about how screen readers are actually used, and it reframes a whole class of markup decisions. A heading chosen for its size and a heading chosen for its level are different decisions, and **styling a div to look like a heading produces a page with no table of contents at all. Everything in this section follows from the jump-by-heading operation**, including why the link-text criterion exists.

AND THE THING TO INTERNALISE IS THAT IT IS A SERIAL CHANNEL WITH A CURSOR

Sight is parallel: you perceive layout, grouping, emphasis, and relative size simultaneously and then attend to one part of it (CSCE 671 Module 02's preattentive material) — the structure arrives before you read anything.

Speech is serial and has a cursor — one thing at a time, in a definite order, with a current position — so **everything your layout conveys spatially has to be conveyed some other way**, because there is no simultaneity to carry it.

Which is why grouping has to be semantic rather than visual: a visual box drawn around three related fields conveys nothing at all serially, and a fieldset with a legend does — the group membership is announced with each field.

And it is why unannounced change is the characteristic defect of modern interfaces — a sighted user sees new content appear in the corner of their eye and a serial user does not know it happened at all — which is **Module 08 §4's live region problem** and the commonest failure in single-page applications.

3 Colour and contrast

COLOUR MUST NEVER BE THE ONLY CHANNEL CARRYING INFORMATION

A substantial minority cannot distinguish certain hue pairs — red and green most commonly, at roughly one in twelve men and one in two hundred women of northern European descent — so a red-green status encoding is simply unreadable to them, and red-green status encoding is everywhere.

So pair colour with a second channel: a shape, an icon, a position, a fill pattern, or a text label — and the text label is the most reliable, because it also works for the screen reader case and for the greyscale print case.

And the quickest test is greyscale: if the information survives desaturation, the colour was redundant — which takes one screenshot and one filter and is the test to run before shipping any figure or status display.

Which is CSCE 679 Module 04's argument — and there it is a perceptual fact about channel reliability and here it is a requirement with a conformance criterion attached, which is the same thing arriving from two directions and is the clearest case of the two courses converging.

What is being measured	Minimum (level AA)	Enhanced (level AAA)
Body text against its background	4.5:1	7:1
Large text — 18pt, or 14pt bold	3:1	4.5:1
Interface components, and the visible indication of their state	3:1	—
Graphics and icons that carry meaning	3:1	—
Purely decorative content, and disabled controls	No requirement	—

These are computed from the relative luminances of the two colours, not judged by eye — so a contrast failure is a measurement rather than an opinion, and that is exactly what makes it arguable in a design review where 'it looks fine to me' otherwise wins. The measured-not-judged property is why contrast requirements actually get met while vaguer ones do not, which is a general lesson about writing requirements and is Module 07 §2's point.

4 Magnification and zoom

- Text must reach 200% without loss of content or function — and 400% with reflow into a single column, which is the criterion that breaks layouts and is the one to test first.
- Reflow means no two-dimensional scrolling — a magnifier user reading a line that requires horizontal scrolling loses their place at the end of every single line, which makes reading impossible rather than merely slow, and is the reason the criterion is written the way it is.

- **And a magnifier shows a small window of the screen, so anything that depends on two distant parts being visible together fails — a tooltip far from its trigger, an error summary at the top of a long form, a status message in the opposite corner.**
- **Plus text has to actually be text — text rendered into an image pixelates on enlargement and cannot be restyled, recoloured, or read aloud — which is why images of text are a separate criterion rather than a style preference.**
- **And respect the user's chosen font size, which means relative units rather than fixed pixels throughout. 400% zoom with reflow breaks more layouts than anything else in this course — and it takes thirty seconds to check, which is the best effort-to-findings ratio available.**

AND THE THREE TESTS, WHICH TAKE ABOUT FIFTEEN MINUTES TOGETHER

Turn on the screen reader and complete one real task without looking at the screen — which tests the no-vision requirement, **and is the one that teaches you most** because the gap between what you assumed was announced and what is announced is large (§2).

Set the browser to 400% and check for reflow, then **turn on operating system magnification and look for distant dependencies** (§4) — two minutes, and it finds layout defects that no other test does.

Run the contrast checker and the greyscale filter (§3) — **which are numeric and take about a minute each**, and give findings nobody can dispute.

Which is three tests, fifteen minutes, and three distinct requirements — and no amount of visual inspection substitutes for any of them, because the whole point is that your vision is not the one being tested (Module 01 §4's shared error).

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
The WAI tutorials on page structure, images, and tables (free)	§2 — the markup that produces correct screen reader behaviour, with examples of both the right and the wrong version.
WebAIM's screen reader keyboard shortcut references (free)	§2's operations — keep this open the first few times you use a screen reader, because the model is not guessable.
Understanding WCAG 1.4.3 and 1.4.11 on contrast (free)	§3's table , with the derivation of the ratios and the reasoning behind the exceptions.
Understanding WCAG 1.4.10 on reflow (free)	§4's hardest criterion — and the explanation of why horizontal scrolling is disqualifying rather than merely awkward.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **State the three vision requirements** and one fix that helps only one of them.
- 2 **List the headings of one of your pages** and ask whether they form an outline.
- 3 **List all the links** out of context and check each makes sense.
- 4 **Complete one task with a screen reader, eyes closed**, and record where you got stuck.
- 5 **Find one visually grouped set of fields** and check whether the grouping is semantic.
- 6 **Find one unannounced change** in an interface you use.
- 7 **Desaturate three interfaces** and list what information disappeared.
- 8 **Measure the contrast** of every text colour in your own project.
- 9 **Set 400% zoom** on your own project and record what breaks.
- 10 **Find a distant dependency** that a magnifier window would split.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name the three vision requirements and the fix for each.
- 2 Why does conflating them leave interfaces half-done?
- 3 Give five screen reader navigation operations.
- 4 Why is heading structure navigation?
- 5 Why must link text work out of context?
- 6 Contrast parallel and serial perception, and give one consequence.
- 7 State the colour rule and the quickest test for it.
- 8 Give the contrast ratios for body text, large text, and components.
- 9 Why does the measured-not-judged property matter?
- 10 What does reflow require, and why is horizontal scrolling disqualifying?